Appendix H

The Extraction Chart: Impedance, Reflection, and the Matching Problem

The electrodynamic formalism developed in Chapter 2 and consolidated in the Equation Registry supplies a resistance R (Equation B.9), a capacitance C (Equation B.10), an inductance L (Equation B.11), a complex wage $W = \psi_m + j\psi_s$ (Equation B.7), and a complex power $S = P_{\text{real}} + jQ_{\text{reactive}}$ (Equation B.12). These quantities are introduced separately and calibrated separately. This appendix supplies the single geometric object on which all of them are simultaneously legible: the **reflection-coefficient plane**, known in electrical engineering as the Smith chart and called here the **Extraction Chart**.

The appendix also closes a definitional gap. Chapter 22 reports a per-axis impedance for the racial channel ($|Z| \approx 0.10$) and per-axis natural frequencies, and uses those numbers to explain the observed 11:1 resonance advantage. No reference impedance Z_0 is defined anywhere in the manuscript, and no normalization is stated. Sections H.2 and H.9 supply both.

Confidence tier. The formal apparatus of this appendix is Tier 3 (structural). It restates and unifies results already derived elsewhere in the book and introduces no new calibrated measurement. The per-axis placements in Section H.9 inherit the Tier 2 status of the spectral decomposition in Chapter 22 for their resistive components, and carry Tier 3 status for their reactive components, which depend on a quality factor Q_k that the present dataset does not resolve. Falsification criteria appear in Section H.10.

H.1 Port Convention

The manuscript describes power flowing in two directions. The Preface establishes that the kinetic labor of $O_{\text{racialized}}$ and I_{buffer} constitutes the actual power supply V_{cc}, with E operating as a gating control layer. Chapters 14 and 22 describe a suppression signal propagating downward from E through P_{uppet} and F_{enforce} into the lower tiers. A transmission-line treatment requires these two directions to be fixed once and used consistently.

Extraction line convention

The extraction architecture is modeled as a two-port network terminated in a load. The **incident wave** a is the suppression signal injected downward from E. The **reflected wave** b is the component returned upward as refusal, backlash, or non-compliance. The **load** is the population upon which the suppression signal terminates. Real power absorbed by the load is the power the population takes on and cannot return, and this absorbed power is the extraction.

This convention is a modeling choice. It is adopted because it makes the reflected wave b the formal object corresponding to organized resistance, which is the quantity the historical chapters track.

H.2 Normalized Impedance and the Reflection Coefficient

Let Z_0 denote the **characteristic impedance of the extraction line**: the impedance a population presents when the suppression signal propagates through it without reflection. Z_0 is set by the carrier established in Chapter 22 and functions throughout as the reference against which all other impedances are measured. Define the normalized impedance of a load:

$$z \;=\; \frac{Z}{Z_0} \;=\; r + jx \tag{H.1}$$

The resistive component r carries the material wage ψ_m and performs real work. The reactive component x carries the psychological wage ψ_s and performs no net work over a cycle. The complex wage of Equation B.7 is therefore an impedance, and $W = \psi_m + j\psi_s$ specifies a position on the chart.

The reflection coefficient is the Möbius transform of the normalized impedance:

$$\Gamma \;=\; \frac{z-1}{z+1}, \qquad z \;=\; \frac{1+\Gamma}{1-\Gamma} \tag{H.2}$$

Equation H.2 maps the entire right half-plane $\mathrm{Re}(z) \geq 0$ onto the closed unit disk $|\Gamma| \leq 1$. Every passive load in the system occupies a point in that disk. The Extraction Chart is that disk.

The power absorbed by the load follows directly:

$$P_{\text{abs}} \;=\; |a|^2\big(1-|\Gamma|^2\big) \tag{H.3}$$

and the Reparations Integral (Equations 0.20 and B.15) admits an equivalent statement in wave variables:

$$\mathcal{R}(t_0, t_{\text{now}}) \;=\; \int_{t_0}^{t_{\text{now}}} \Big(|a(\tau)|^2 - |b(\tau)|^2\Big)\, d\tau \tag{H.4}$$

Accumulated extraction is accumulated absorbed power: the portion of the incident suppression signal that the population took on and did not return.

H.2.1 The Enclosure Score as a Match Condition

The Tri-Modal Enclosure Model (Equation 0.5) measures the degree to which an Out-group is denied all practical routes of escape, resilience, and structural self-perception, with $\mathcal{S}_{\text{enc}} \to 1.0$ denoting absolute subjugation. A population with no route of return reflects nothing:

$$\mathcal{S}_{\text{enc}} \;=\; 1 - |\Gamma| \tag{H.5}$$

Total enclosure is a perfect impedance match. The center of the Extraction Chart, $\Gamma = 0$, $z = 1$, $Z = Z_0$, is the condition the extraction algorithm optimizes toward. The rim, $|\Gamma| = 1$, is withdrawn compliance: a purely reactive load that absorbs no real power. The Preface states that the architecture dies when compliance is withdrawn. Equation H.3 is the formal content of that statement.

The standing-wave ratio provides the system's diagnostic readout:

$$\text{VSWR} \;=\; \frac{1+|\Gamma|}{1-|\Gamma|} \tag{H.6}$$

Visible unrest is a standing-wave measurement. VSWR diverges as the load approaches the rim and equals unity at perfect enclosure.

H.3 The Buffer Class as a Lossless Matching Network

The Buffer Work Theorem (Equation B.14) establishes that the magnetic component of the extraction force performs zero net work on I_{buffer}:

$$W^{\text{psych}}_{I_{\text{buffer}}}(t_0, t_1) = \int_{t_0}^{t_1} \mathbf{Q}\,(\vec{v} \times \vec{B}) \cdot \vec{v}\, d\tau \;\equiv\; 0$$

Zero net work over a cycle with nonzero instantaneous magnitude is the defining property of a reactive element. In network terms, an element that dissipates nothing, stores and returns energy each cycle, and is inserted between a source and a load in order to eliminate reflection is a **matching network**.

Theorem H.1 (Buffer Matching Theorem)**.** *I_{buffer} occupies the structural position of a lossless impedance-matching network between E and $O_{racialized}$. Its compensation is purely reactive: the time-averaged real power delivered to I_{buffer} is zero, while the reactive power $Q_{reactive}$ delivered to it is nonzero and sustained.*

Argument. By Equation B.14 the psychological wage transfers no net energy. By Equation B.12 the complex power delivered to any element decomposes as $S = P_{\text{real}} + jQ_{\text{reactive}}$, with P_{real} the time-averaged dissipation. An element with $P_{\text{real}} = 0$ and $Q_{\text{reactive}} \neq 0$ is purely reactive. A purely reactive element placed in series or shunt between source and load transforms the impedance the source sees without dissipating power, which is the definition of a lossless matching network. The Buffer Class is positioned between E and $O_{\text{racialized}}$ in the five-tier topology by construction (Chapter 4). □

Theorem H.1 resolves a question the historical chapters raise repeatedly and answer only descriptively: why the Buffer Class defends a partition from which it accumulates little. Reactive power has nonzero magnitude and zero time-average. The compensation is received in full and is worth nothing at the end of each cycle. A matching network holds its position because $|Q_{\text{reactive}}|$ is large, and it accumulates nothing because P_{real} is zero.

On the chart, a lossless matching element moves a load along a constant-r circle (series reactance) or a constant-g circle (shunt susceptance) without changing the real part. The trajectory of I_{buffer} across the historical record is a trajectory along such circles.

H.4 Reform as Impedance Matching

Chapter 19 argues that reform serves the extraction algorithm. The chart converts that argument into a monotonicity result.

Theorem H.2 (Reform Monotonicity)**.** *Any intervention that strictly reduces $|\Gamma|$ strictly increases the real power absorbed by the load. Formally, from Equation H.3,*

$$\frac{\partial P_{abs}}{\partial |\Gamma|} \;=\; -2\,|a|^2\,|\Gamma| \;<\; 0 \qquad \textit{for } |\Gamma| > 0 \tag{H.7}$$

Reforms that reduce friction, reduce backlash, and reduce refusal reduce $|\Gamma|$ by construction. Reflection and extraction are the same variable carrying opposite signs, so an intervention cannot reduce both. Reform operates as a matching stub.

The corollary specifies the direction of any anti-extractive intervention: increase $|\Gamma|$, move outward on the chart toward the rim, and terminate in a purely reactive load. This is deliberate mismatch, and it is the formal content of withdrawn compliance.

H.5 Rotation, the Electoral Clock, and Quarter-Wave Co-optation

On a lossless line, displacement of the reference plane rotates the reflection coefficient without changing its magnitude:

$$\Gamma(\ell) \;=\; \Gamma_L\, e^{-2j\beta\ell}, \qquad \beta = \frac{2\pi}{\lambda} \tag{H.8}$$

One full rotation of the chart corresponds to $\ell = \lambda/2$. Chapter 22 establishes the four-year presidential cycle as the system's dominant spectral mode, with a 24:1 power advantage for identity-band language at $f = 0.25$ cyc/yr. Setting λ equal to the electoral cycle makes Equation H.8 a clock.

Theorem H.3 (Quarter-Wave Co-optation)**.** *Let a political formation present the extraction line with a short circuit, $z_L = 0$, $\Gamma_L = -1$: total refusal, drawing the full suppression current and returning it inverted. After a displacement of $\ell = \lambda/4$, the impedance presented at the new reference plane is*

$$z_{in} \;=\; \frac{1}{z_L} \;\longrightarrow\; \infty, \qquad \Gamma_{in} = +1 \tag{H.9}$$

which is an open circuit: a formation that draws no current and participates in nothing. The composition of the load is unchanged throughout.

The transformation requires no change in the formation's membership, program, or intensity. Its electrical distance from the generator changes, and the same load presents the system its exact inverse. A further $\lambda/4$ returns it to a short. Militancy, irrelevance, and militancy recur with a period the carrier fixes.

Theorem H.3 yields a testable prediction: insurgent formations should exhibit periodicity phase-locked to the electoral carrier, detectable by the same spectral method Chapter 22 applies to congressional language. The time-domain test that failed in that

chapter ($t = 0.479$, $p = 0.634$) failed for the reason stated there, namely that phase-locked oscillations with variable amplitude are invisible to tests that ignore phase. The same methodological point applies here.

H.5.1 Attenuation and Manufactured Consent

On a lossy line the reflection coefficient spirals inward toward the center with distance:

$$\Gamma(\ell) \;=\; \Gamma_L \, e^{-2\alpha\ell} \, e^{-2j\beta\ell} \tag{H.10}$$

An observer positioned at electrical distance ℓ from the load measures $|\Gamma|$ reduced by $e^{-2\alpha\ell}$ and reads the system as matched. Attenuation manufactures the appearance of consent. Historical distance from the site of extraction produces apparent settlement as an artifact of the line, and the load itself is unchanged.

H.6 Non-Reciprocity: The Enforcement Class as an Isolator

The lethal-autonomy gradient (Equation 9.1) states

$$\text{Lethal Autonomy}(F_{\text{enforce}}) \gg \text{Lethal Autonomy}(I_{\text{buffer}}) \gg \text{Lethal Autonomy}(O) = 0$$

In scattering parameters this is a statement about a two-port network:

$$|S_{21}| \;\gg\; |S_{12}| \;\approx\; 0 \tag{H.11}$$

Force passes downward through F_{enforce} and does not pass back up. A device with this property is an **isolator**.

Lorentz reciprocity holds for any linear medium whose permittivity and permeability tensors are symmetric, and it forces $S_{12} = S_{21}$. Violating reciprocity requires nonlinearity, time-variance, or a static magnetic bias that renders the permeability tensor antisymmetric. The standard engineering realization is a ferrite held in an applied DC magnetic field. The bias field performs no work on the signal and is the sole reason the device conducts in one direction.

Theorem H.4 (Cultural Bias and Non-Reciprocity)**.** *The cultural magnetic field $\vec{B}$ defined in Section 0.2 functions as the static bias that breaks reciprocity in the enforcement tier.*

Removing the bias restores $S_{12} = S_{21}$, under which force applied downward can propagate upward with equal transmission.

Theorem H.4 supplies the load-bearing role of a field that performs no work. The framework establishes in Equation B.14 that $\vec{B}$ transfers no energy, and the Tri-Modal Enclosure Model assigns the psychological mode e_3 to it. The isolator identification explains why a channel that transfers zero energy is structurally indispensable: the asymmetry of the enforcement tier exists only under magnetic bias, and the enforcement tier becomes a symmetric two-way channel without it.

H.7 The Active Region and the Counter-Virus

Every passive load satisfies $\mathrm{Re}(z) \geq 0$ and therefore $|\Gamma| \leq 1$. The condition

$$|\Gamma| > 1 \quad \Longleftrightarrow \quad \mathrm{Re}(z) < 0 \tag{H.12}$$

requires negative resistance: an element that returns more power than it receives. Equation H.12 is also the oscillation condition. A network terminated in a negative resistance grows without bound at the frequency where the loop gain exceeds unity.

Chapter 21 calls for a counter-virus. Equation H.12 defines it. Every point inside the unit disk is a passive load absorbing some fraction of the incident suppression signal, and repositioning within the disk changes that fraction. Leaving the disk requires the population to function as a source. This is consistent with the resonance result of Chapter 1, where sustained coordination at the natural frequency $\omega_0 = 1/\sqrt{LC}$ drives the response of a low-resistance system without bound.

The compactification is itself a result. Equation H.2 maps an infinite half-plane onto a bounded disk with a fixed boundary. Appendix G conjectures that the topology of power is finite. The Extraction Chart is the statement that unlimited variation in impedance occupies a bounded space with a fixed edge.

H.8 The Bode–Fano Limit on Simultaneous Axes

The chart is defined at a single frequency. The framework is explicitly multi-band: Chapter 22 resolves race, gender, and sexuality into sub-bands with distinct natural frequencies. The Bode–Fano criterion bounds the match a reactive load admits across a finite bandwidth. For a load reducible to a parallel RC combination,

$$\int_0^\infty \ln \frac{1}{|\Gamma(\omega)|}\, d\omega \;\le\; \frac{\pi}{RC} \tag{H.13}$$

The integral is a fixed budget. Improving the match over one band requires accepting a worse match over another. Equation H.13 is a quantitative statement of the cannibalization argument in Chapter 16: the system cannot hold every axis matched at once, and tightening enclosure on one axis necessarily loosens it elsewhere in the spectrum. The historical pattern of sequential axis activation follows from a conserved matching budget.

H.9 Per-Axis Placement

Chapter 22 reports exploratory dominant spectral periods recovered by Lomb–Scargle from the selected SCOTUS corpus: race at 3.6 yr and gender at 6.2 yr. The sexuality period is unresolved. Modeling each axis as a series resonant branch driven at the four-year carrier, the normalized branch impedance is

$$z_k(\omega) \;=\; r_k\Big[1 + jQ_k\,\delta_k\Big], \qquad \delta_k \;=\; \frac{\omega}{\omega_{0,k}} - \frac{\omega_{0,k}}{\omega} \tag{H.14}$$

The race and gender detuning values follow from those exploratory period estimates. The sexuality row remains unresolved:

Axis	Dominant period	$\omega/\omega_{0,k}$	δ_k
Race	3.6 yr	0.900	-0.211
Gender	6.2 yr	1.550	$+0.905$
Sexuality	threshold-activated	—	—

The sign of δ_k is the robust result. Race is driven below its natural frequency and presents a capacitive reactance, placing it below the real axis. Gender is driven above its natural frequency and presents an inductive reactance, placing it well above the real axis and far from match. The magnitude of each reactance requires the quality factor Q_k, which the present dataset does not resolve; the placements in Figure H.1 use $Q_k = 3$ as an illustrative normalization and are Tier 3 in their reactive component.

H.9.1 Phase Loading as Current Division

The identification of the axes as parallel branches supplies a circuit-theoretic definition for the phase-loading coefficients ρ_k that appear in the Spectral Carrier (Equation B.13), the Unified Lorentz Force (Equation B.6), and the Reparations Integral (Equation 0.20):

$$\rho_k = \frac{Y_k}{\sum_{i=1}^{N} Y_i}, \qquad Y_k = \frac{1}{Z_k} \tag{H.15}$$

The suppression current divides among the available axes in proportion to branch admittance. The axis with the lowest impedance carries the largest share. Race carries $|Z| \approx 0.10$, the lowest of the three, and receives the largest ρ_k, which is the same claim as the 11:1 resonance advantage expressed as a current-division ratio. Equation H.15 converts ρ_k from an assigned weight into a derived quantity.

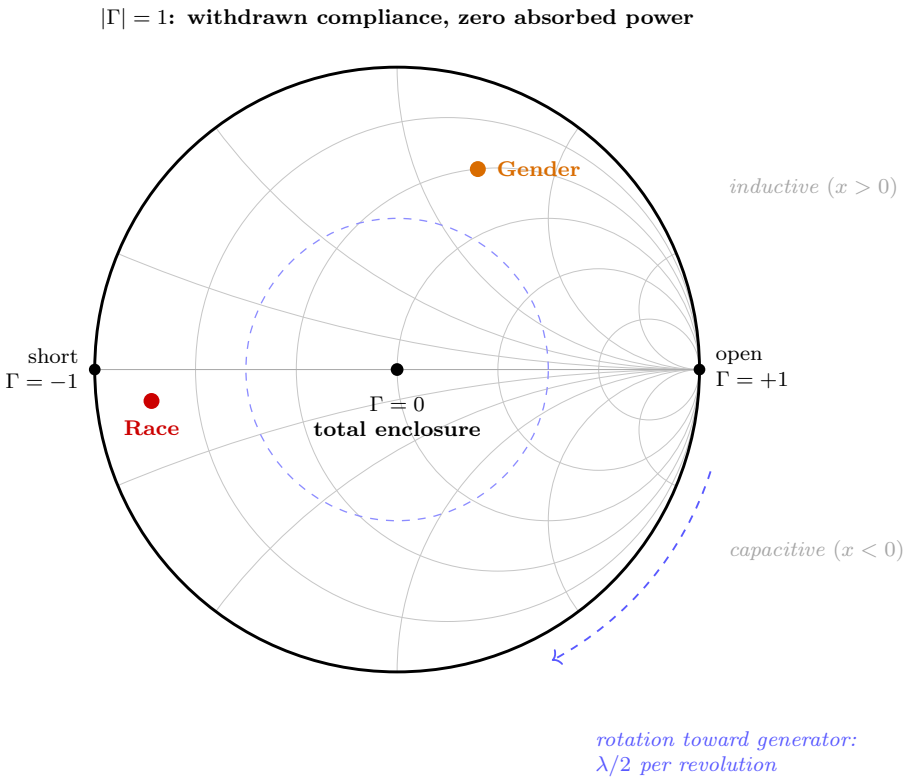


Figure H.1: **The Extraction Chart.** The reflection-coefficient plane of the extraction line. Every passive load occupies a point in the unit disk. The center ($\Gamma = 0$, $z = 1$) is a perfect impedance match and corresponds to total enclosure, $\mathcal{S}_{\text{enc}} \to 1.0$, with all incident suppression absorbed and nothing returned. The rim ($|\Gamma| = 1$) is withdrawn compliance: a purely reactive termination absorbing zero real power. Grey circles are loci of constant normalized resistance r (material wage) and constant normalized reactance x (psychological wage). Displacement along the line rotates a load clockwise about the center at one revolution per $\lambda/2$, which places a short circuit at the open-circuit point after $\lambda/4$ (Theorem H.3). Coloured markers place the race and gender sub-bands using exploratory dominant periods recovered by Lomb–Scargle from the selected SCOTUS corpus. Sexuality has no resolved period and is unplotted. Equation H.14 sets the reactive component using an illustrative $Q_k = 3$, which carries Tier 3 status.

H.10 Falsification Criteria

The claims of this appendix fail under the following conditions.

1. **Buffer Matching Theorem.** A longitudinal dataset showing that I_{buffer} accumulates real material wealth attributable to the psychological wage, net of its own labor contribution, over a period exceeding one generation. Nonzero time-averaged P_{real} delivered to the Buffer Class falsifies Theorem H.1.

2. **Reform Monotonicity.** A documented intervention that measurably reduced backlash, refusal, and non-compliance while simultaneously reducing the extraction rate $d\mathcal{E}/dt$. Equation H.7 forbids this pairing.

3. **Quarter-Wave Co-optation.** Spectral analysis of insurgent-formation activity showing no power concentration at the electoral carrier or its harmonics across a dataset of comparable length to the 60-year congressional series. Absence of phase-locking falsifies Theorem H.3.

4. **Cultural Bias and Non-Reciprocity.** A documented enforcement architecture exhibiting the lethal-autonomy gradient of Equation 9.1 in the absence of any cultural or psychological partition. Sustained non-reciprocity without bias falsifies Theorem H.4.

5. **Bode–Fano Limit.** A historical period in which enclosure tightened simultaneously and durably across all identity axes without compensating loosening on any axis. Equation H.13 forbids an unbounded matching budget.

6. **Phase loading as current division.** Measured ρ_k values departing from the admittance ratio of Equation H.15 beyond the uncertainty of the per-axis impedance estimates.

H.11 Open Problems

Three quantities in this appendix carry assumed values, and each is resolvable with existing data.

Z_0 is defined structurally in Section H.2 and awaits a fitted value. Estimating it from the congressional-record spectral series would place the per-axis impedances on an absolute scale and upgrade the placements in Figure H.1 from Tier 3 to Tier 2.

The quality factors Q_k control the reactive magnitudes and are currently set to an illustrative constant. Resolving them requires higher-frequency data than the annual series supports, which is the same resolution limit that left the two-year midterm cycle indeterminate in Chapter 22.

The Bode–Fano budget in Equation H.13 has not been evaluated numerically. A computed bound would convert the cannibalization argument from an ordinal claim into a stated maximum on the number of axes the architecture can hold matched at once.